AM41RS Specialist Research Skills and Techniques
CS4780 Research Methods and Professional Practice

Dissertation Topic

to be addressed by students who have not secured
their own individual dissertation topic
by 10 March 2025

Please note that this is NOT the project topic that you will be working on, but is offered solely for the purpose of the research proposal assignment within this module. You will be allocated your individual dissertation topic by your programme director/dissertation module coordinator.

The UK Energy Act 2023 led to the establishment of the National Energy System Operator (NESO), as an independent, public corporation planning and operating the whole energy system in Great Britain. Through ensuring that the energy supply meets demand they make sure that the country has the essential energy it needs. They have an Open Data Portal, where they bring together from multiple sources and make publicly available numerous datasets about the energy system, including energy demand, generation and carbon intensity.

"Harnessing the power of AI to unlock invaluable insights, to make critical evidence-based decisions and ultimately drive tangible benefits for consumers" is listed as one of NESO's priorities (https://www.neso.energy/what-we-do/data-ai).

In this open-ended project you are expected to devise a FINER research question (https://scientific-publishing.webshop.elsevier.com/research-process/finer-research-framework/) that may be of value to NESO and address this question using data science, machine learning and other artificial intelligence methods that you studied on your MSc programme. There is no restriction, whether to focus on energy consumption prediction, demand prediction or any other aspect.

As a reminder, the characteristics of FINER questions are explained below:
Feasible: The question must be answerable within the timeframe and scope of the MSc dissertation.
Interesting: The question must be of interest to others, such as NESO and members of the public.
Novel: The novelty can be a new way of applying a known method, a new comparative analysis of known methods or a method that involves some new elements.
Ethical: You must convince that the proposed approach is ethical.
Relevant: The question must have real world relevance.

For the **dissertation proposal**, you must investigate the context, conduct a brief initial literature review, formulate the research question and plan the methodology to answer the research question, so that you convince that, should you have the opportunity to conduct the project, by the end of it you are likely to offer valuable insight to NESO.